openMPPT Hardware Design Calculations

Theoretical basis and calculations for component selection

1. Power Stage Parameters

Parameter	Symbol	Value	Notes
Switching Frequency	f_{sw}	100 kHz	Defined in `mppt.h` (`TIMER_PERIOD = 240`)
Maximum Input Voltage	$V_{\mathrm{in,max}}$	80 V	Hardware limit
Maximum Output Current	$I_{\mathrm{out,max}}$	20 A	Safety limit
Nominal System Voltage	V_{sys}	12V / 24V / 48V	Target batteries
Gate Drive Voltage	V_{drive}	10 V	Optimized for IRS21867 thermals (v1.3)

Planned change: 100kHz → 200kHz main switching frequency. Not yet implemented — `TIMER_PERIOD` in `mppt.h` (currently 240) and the table above still reflect 100kHz. `setup_cubemx_env_auto.py` already derives `TIMER_PERIOD` from `MPPT.ioc` and `main.c`'s PWM limits/dead-bands are dynamic, so the firmware side should follow from changing the timer config. See Section 8 for the full component re-check.

2. Gate Drive & Auxiliary Power Optimization (v1.3)

Primary Aux Supply (10V) - SCT2A25: Replaces the unstable 12V XL7005A with a robust 100V-rated step-down (SCT2A25).

- Output is tuned to **10V** instead of 12V.
- **Feedback Divider**: $V_{\mathrm{FB}} = 1.2\mathrm{V}$
  - $V_{\mathrm{OUT}} = 1.2 \times \left(1 + \frac{R_{\mathrm{up}}}{R_{\mathrm{down}}}\right)$
  - Using E96 series: $R_{\mathrm{up}} = 110\ \mathrm{k}\Omega$, $R_{\mathrm{down}} = 15\ \mathrm{k}\Omega \rightarrow V_{\mathrm{OUT}} = 1.2 \times \left(1 + \frac{110}{15}\right) = 10.0\mathrm{V}$
- **Switching Frequency**: Fixed at 300kHz.
- **Inductor** (L_1): For 10V out, a standard **33μH or 47μH** inductor (e.g., 6x6mm shielded) is suitable.
- **UVLO Divider**: To set start at 12.5V: $R_{\mathrm{UVLO\_TOP}} = 430\ \mathrm{k}\Omega$, $R_{\mathrm{UVLO\_BOT}} = 47\ \mathrm{k}\Omega$ (Both are E24).
- **SW-node Freewheeling Diode**: D7, **SS510B** (100V, 5A Schottky, C7420368) between the switch node and GND. (Previously documented here as "US1M" — never actually placed; verified against the schematic's real net connections.)

Secondary Aux Supply (3.3V Logic) - SY8120:

- Uses a "Cascaded Buck" architecture: 10V → Tiny Sync Buck (SY8120) → 3.3V.
- **Feedback Divider**: $V_{\mathrm{FB}} = 0.6\mathrm{V}$
  - $V_{\mathrm{OUT}} = 0.6 \times \left(1 + \frac{R_{\mathrm{up}}}{R_{\mathrm{down}}}\right)$
  - Using E24 series: $R_{\mathrm{up}} = 68\ \mathrm{k}\Omega$, $R_{\mathrm{down}} = 15\ \mathrm{k}\Omega \rightarrow V_{\mathrm{OUT}} = 0.6 \times \left(1 + \frac{68}{15}\right) = 3.32\mathrm{V} \approx 3.3\mathrm{V}$
- **Switching Frequency**: Fixed at 500kHz.
- **Inductor** (L): **4.7μH** (recommended for 3.3V out).

3. Transient Voltage Suppression (TVS) Strategy

Primary Bus Protection (80V):

- **Component**: **5.0SMDJ85CA** (Bidirectional, 5kW).
- **Rationale**: Standoff (V_{RWM}) of 85V ensures no leakage at 80V. 5kW rating handles massive inductive kickback and solar surges.

- **Placement**: Immediately adjacent to VIN/VOUT XT60 connectors.

Gate Drive Protection (10V):
- **Component**: **H12VH22U** (6kW Surge).
- **Rationale**: 12V standoff ensures invisibility at 10V rail. Protects sensitive IRS21867 drivers (25V max) from regulator failure.

Logic Protection (3.3V):
- **Component**: **H3V3L06B** (ESD).
- **Rationale**: 3.3V standoff is required because STM32 Absolute Max is only 4.0V. Provides last line of defense against ESD and noise.

4. Inductor Sizing (Main Power Stage)

Targeting a peak-to-peak ripple current (ΔI_L) of 20% of $I_{\text{out,max}}$ (**4.0 A**).

Buck Mode Worst-Case: (50% Duty Cycle, e.g., 80V in, 40V out)

$$L_{\text{buck}} = \frac{V_{\text{out}} \times (V_{\text{in}} - V_{\text{out}})}{\Delta I_L \times f_{\text{sw}} \times V_{\text{in}}}$$

At $f_{\text{sw}} = 100\text{kHz}$: $L_{\text{buck}} = \frac{40\times40}{4.0\times100000\times80} = * 50.0\mu\text{H} *$

v1.3 Component Selection: Target inductor size: **33 µH to 47 µH**.
- **CRITICAL**: Saturation Current (I_{sat}) **MUST be $>$ 25A**. Using a lower I_{sat} (like 11A) will cause magnetic collapse and MOSFET destruction.

5. Bulk Capacitor Selection

Targeting an output voltage ripple (ΔV_{out}) of $<$ **100mV**.

$$\text{ESR}_{\text{max}} = \frac{\Delta V_{\text{out}}}{\Delta I_L} = \frac{0.1\text{V}}{4.0\text{A}} = * 25m\Omega *$$

v1.3 Capacitor Strategy:
- Use **4x 330µF 100V Low-ESR Electrolytic** (e.g., Ymin LKML series) in parallel.
- **Total Bank ESR**: $\approx 11.75m\Omega$ (Passes 25mΩ limit).
- **Total Ripple Capacity**: $\approx$ 8.5 A (Safe for 20A operation).

6. Voltage Divider & ADC Scaling

Voltage Sensing LPF (v1.3 Optimized):
- $R_{\text{top}} = 200$ kΩ, $R_{\text{bottom}} = 4.7$ kΩ.
- $R_{\text{th}} \approx 4.59$ kΩ
- $C_{\text{filter}} = 10$ nF
- $f_c = \frac{1}{2\pi\times4590\times10\times10^{-9}} \approx 3.47$ kHz
- *Evaluation*: Balanced for high-speed tracking and noise rejection.

7. Current Sense (CC6937S8-3FB020, U2/U3)

Verified against the CrossChip `CC6937` datasheet (not a name-convention guess).

- **Type**: Isolated Hall Effect, 3750V_{RMS} isolation (pins 1-4 to 5-8) — 47x margin over the 80V bus.
- **Sensitivity**: **66mV/A** (confirmed from the ordering table for the −3FB020 variant).
- **Supply**: VCC 3.0-3.3V nominal — the “3” in “3FB” denotes the 3.3V-supply series, matching the system’s logic rail exactly.
- **Range**: ±20A. Zero-current output $V_{\text{OUTQ}} \approx 1.65$V (VCC/2, confirmed), full scale swings to $\approx$ 0.33-2.97V — centered and within a 3.3V ADC’s linear range.

- **Response time**: 1.5µs (10-90% step) — well under one 200kHz switching period (5µs), adequate for fast overcurrent/backflow trip.
- **Bandwidth**: 230kHz −3dB, 150x margin over the 1.5kHz control loop sample rate. Margin over switching frequency shrinks from 2.3x (100kHz) to 1.15x (200kHz) at full power — not a concern for this design (average sensing + fast-trip response time, not raw bandwidth, govern both use cases here), but worth knowing if the sensor's role ever changes to reconstructing switching-ripple waveforms directly.
- **Insertion loss**: 0.5mΩ internal conductor resistance — negligible, and the reason this replaced the INA240+shunt approach from v1.1 (per `ROADMAP_V1.3.md`).
- **Decoupling**: **1nF** VREF-to-GND (matches the datasheet's own `CREF=1nF` test condition; confirmed in the schematic as C10/C26) and a separate **100nF (0.1µF)** VCC bypass (confirmed as C12 near U2). Also 1nF at VOUT per the datasheet's `COUT=1nF` condition.

8. 200kHz Migration: Component Re-Check

This section re-verifies the parts already selected for 100kHz operation against a doubled switching frequency, using values pulled directly from the vendored datasheets (not assumed). Nothing here has been implemented in firmware or hardware yet — this is the pre-check called for by the `STANDARDS.md` "Part selection & calculations" mandate before the `ROADMAP_V1.3.md` Phase 4 checklist item is closed.

8.1. Main Switching MOSFETs (Q1-Q4, BSC030N08NS5, C501507)

Source: Infineon `BSC030N08NS5` datasheet, Rev. 2.3, 2019-10-31.

Parameter	Value	Condition
$R_{\mathrm{DS(on)}}$	2.6 mΩ typ / 3.0 mΩ max	V_{GS}=10V, I_D=50A
Q_g (total gate charge)	61 nC typ / 76 nC max	V_{DD}=40V, I_D=50A, V_{GS}=0→10V
t_r (turn-on rise)	12 ns typ	V_{DD}=40V, V_{GS}=10V, I_D=50A, $R_{\mathrm{G,ext}}$ =3Ω
t_f (turn-off fall)	13 ns typ	same
C_{oss}	700 pF typ / 910 pF max	V_{DS}=40V, f=1MHz
$R_{\mathrm{th\ JA}}$ (PCB only, no heatsink)	50 K/W	6 cm² copper, 1-layer, natural convection
$T_{\mathrm{j,max}}$	150 °C	—

Conduction loss (frequency-independent):

$$P_{\mathrm{cond}} = I_{\mathrm{rms}}^2 \times R_{\mathrm{DS(on),max}} = 20^2 \times 0.0030 \approx * 1.2 \text{ W} *$$

Switching loss (linear-approximation model, worst case $V_{\mathrm{DS}} = 80\text{V}$, $I_D = 20\text{A}$):

$$P_{\mathrm{sw}} = \frac{1}{2} \times V_{\mathrm{DS}} \times I_D \times \left(t_r + t_f\right) \times f_{\mathrm{sw}}$$

	100 kHz (current)	200 kHz (proposed)
P_{sw} per hard-switched MOSFET	2.0 W	4.0 W
P_{cond} per MOSFET	1.2 W	1.2 W (unchanged)
Total est. loss	**≈3.2 W**	**≈5.2 W**

Thermal flag. With PCB-only cooling ($R_{\mathrm{th\ JA}}$=50 K/W, no heatsink/thermal vias), the datasheet's own max package dissipation at T_A=25°C is $P_{\mathrm{tot}} = \frac{150-25}{50} = * 2.5 \text{ W} *$ — already below the ≈3.2W estimate at **100kHz**, let alone ≈5.2W at 200kHz. This assumes worst-case V_{DS}/

I_D coincidence and a simplified switching-loss model (real loss depends on which of Q1-Q4 is hard-switched vs. synchronous-rectifying in the actual buck-boost topology, and gate resistance/ drive strength affect t_r/t_f in-circuit). This is exactly what `ROADMAP_V1.3.md` Phase 3′s open item ("20A Thermal Design: widen high-power traces, implement massive thermal via grid") already anticipates — that item should land **before** or **alongside** the 200kHz switch, not after. The bottom-side $R_{\text{th JC}}$ path (0.5-0.9 K/W typ/max) is dramatically better than the PCB-only path and is what a proper thermal via array would unlock.

Gate drive loss (in the driver/gate resistor, not the MOSFET junction — informational):

$$P_{\text{gate}} = Q_g \times V_{\text{GS}} \times f_{\text{sw}} = 76 \times 10^{-9} \times 10 \times f_{\text{sw}}$$

At 100kHz: 76mW/MOSFET. At 200kHz: 152mW/MOSFET. Not a concern for the driver IC's thermal budget either way.

8.2. Gate Driver (IRS21867STRPBF, C52290)

Source: Infineon `IRS21867S` datasheet v01.00.

Parameter	Value	Condition
t_{on} / t_{off} propagation delay	170 ns typ / 250 ns max	$V_{\text{CC}} = V_{\text{BS}}$=15V, C_L=1000pF
MT (delay matching, $\lvert t_{\text{on}} - t_{\text{off}} \rvert$)	35 ns max	—
t_r / t_f (driver output)	22/18 ns typ, 38/30 ns max	—
Output source/sink current	4.0 A typ	—

At 200kHz the switching period is 5.0µs vs. 10µs at 100kHz. The 170-250ns propagation delay grows from 2.5% to 5% of the period — worth including explicitly in dead-time budgeting, but well within normal margins; **not** a hard blocker on 200kHz operation. Output drive current (4.0A) is unchanged and still comfortably swings Q_g=76nC in the tens-of-ns range required. No datasheet-stated maximum switching frequency applies here.

8.3. Main Inductor (L4)

Re-deriving Section 4′s formula at 200kHz:

$$L_{\text{buck}}(200\text{kHz}) = \frac{40 \times 40}{4.0 \times 200000 \times 80} = *\,25.0\mu\text{H}\,*$$

This halves the 100kHz target (50µH → 25µH). The originally-placed part, `FC-SE2822-150M` (C46553544, verified via datasheet, not the name-convention guess this section originally used): **15µH ±20%, Isat 30.5A, rated 30A, DCR 3.5mΩ, 28x23mm THT, $2.30 / 407 in stock**.

	100kHz (current firmware)	200kHz (planned)
Ripple current ΔI_L	13.3A	6.67A
Peak current / I_{sat} margin	26.7A / 14%	23.3A / 31%
Output ripple ΔV_{out} (bank ESR 11.75mΩ)	**157mV — 57% over the 100mV target**	78mV ✓
Conduction loss ($I_{\text{out}}^2 \times$ DCR)	1.40W	1.40W (freq-independent)

L4 only meets the ripple spec at 200kHz — a real, accepted dependency, not a bug. It was deliberately sized on the assumption that the 200kHz switch happens, specifically to avoid the "humongous" 50µH part the 100kHz target would otherwise require. This is not something to fix by re-checking values; it's a documented design decision.

Candidate replacements evaluated — two rounds. First pass (22-30µH / Isat≥30A, filtered to in-stock — 19 of 21 candidates found were out of stock and excluded) optimized purely for margin and found `C37634008` (RSEQ32-220M, Ruishen): 22µH, Isat 90A, DCR 0.9mΩ, $6.26/26 in stock — 304% Isat margin and 0.36W conduction loss at 200kHz, a huge improvement. **Rejected after a second look**: this compared candidates against each other without ever checking the original part's own price/stock ($2.30, 407 in stock) — a 2.7x price and 15x stock gap that makes "maximum margin" the wrong optimization target for a prototype build.

Second pass widened the inductance range (15-33µH) and weighed cost/stock properly:

	FC-SE2822-150M (orig.)	C54830921	**HCZH-3218-150-M (chosen)**	C37634008
Manufacturer	FANGCHENG	Coilank	YOUCHI	Ruishen
Stock / Price	407 / $2.30	69 / $2.87	88 / $3.86	26 / $6.26
L / Isat / DCR	15µH / 30.5A / 3.5mΩ	15µH / 44A / 3.0mΩ	15µH / 30A / 2.2mΩ	22µH / 90A / 0.9mΩ
Package	THT 28x23mm	THT 33.5x28mm	THT 33.5x28mm	THT 33x31.5mm
200kHz ripple / Isat margin	78mV / 31%	78mV / 89%	78mV / 29%	53.4mV / 304%
200kHz conduction loss	1.40W	1.20W	**0.88W**	0.36W

`C54830921` is the best pure value play (+$0.57 for 89% Isat margin, same ripple). `HCZH-3218-150-M` was chosen instead for its lower DCR (0.88W conduction loss, a real efficiency win) at a still-modest $1.56 premium over the original part — a deliberate prototype-stage trade-off (moderate cost increase for a meaningful efficiency gain, not chasing the maximum-margin `C37634008` pick, which cost 2.7x more for headroom this design doesn't need).

Chosen: `HCZH-3218-150-M` (YOUCHI, `C53699952`). Applied to `buck_boost.kicad_sch`; PCB footprint not yet updated — see `STANDARDS.md`'s L4 note for status.

8.4. Summary

Item	**Verdict at 200kHz**
MOSFETs (BSC030N08NS5)	Electrically fine (Vds/Rds(on)/Qg headroom unchanged), but switching loss ≈doubles — thermal design (ROADMAP Phase 3) must land first
Gate driver (IRS21867STRPBF)	No concern — propagation delay and drive current have ample margin
Main inductor (L4)	Replacement chosen: `HCZH-3218-150-M` (C53699952) — clears ripple target at 200kHz, DCR cut from 3.5mΩ to 2.2mΩ (0.88W conduction loss), for a modest $1.56 premium over the original part
Bulk caps /	Frequency-independent, no re-check needed

Item	**Verdict at 200kHz**
TVS / voltage sensing	